

ECSE-4040

Digital Electronics

Lecture 26

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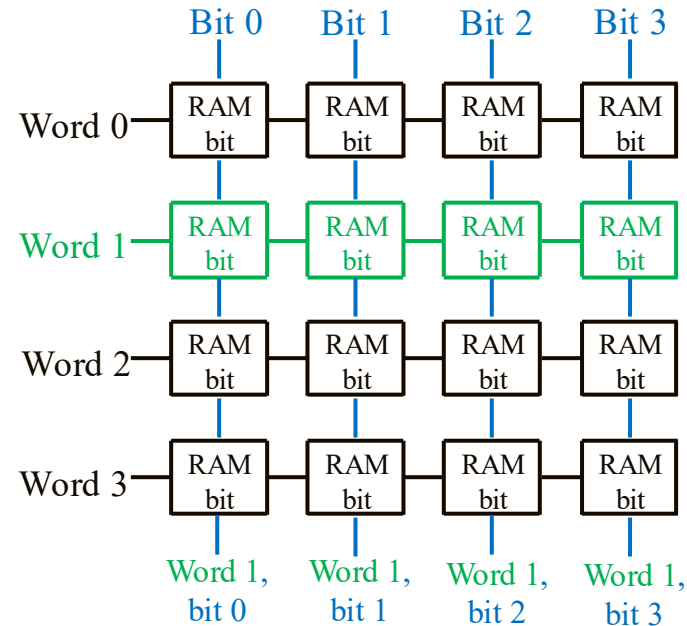
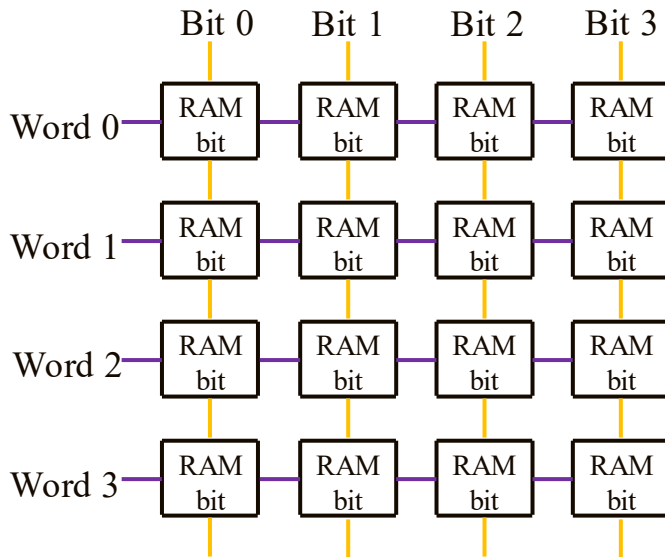
Lecture Outline

- RAM
 - SRAM Operation
 - Pre-charging
 - Multi-port
 - DRAM

RAM Overview

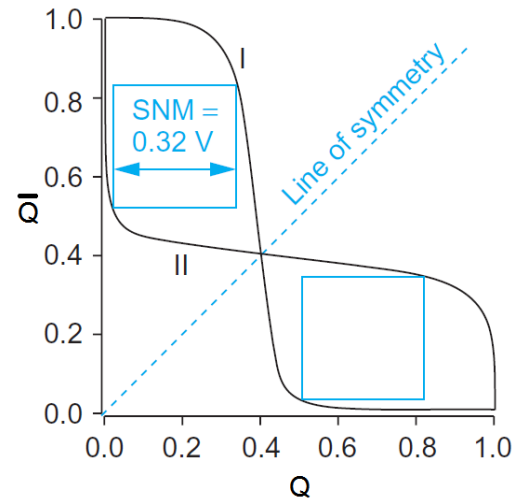
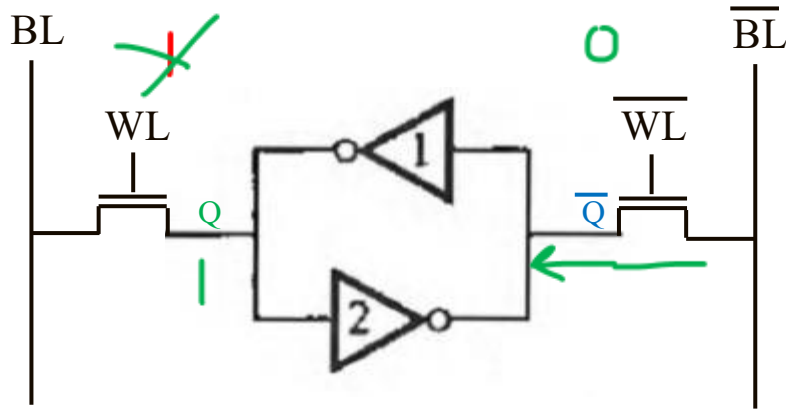
- Need to store data created by logic
- This data can be randomly accessed at any time, dependent on program requirements.
- Two families of RAM
 - Static (SRAM)
 - Storage element created by dual inverter loop.
 - 6 to 8 transistors (4 for basic storage element)
 - Dynamic (DRAM)
 - Storage element is a MOS capacitor (NMOS without S or D connections)
 - 1 transistor total

RAM Arrays

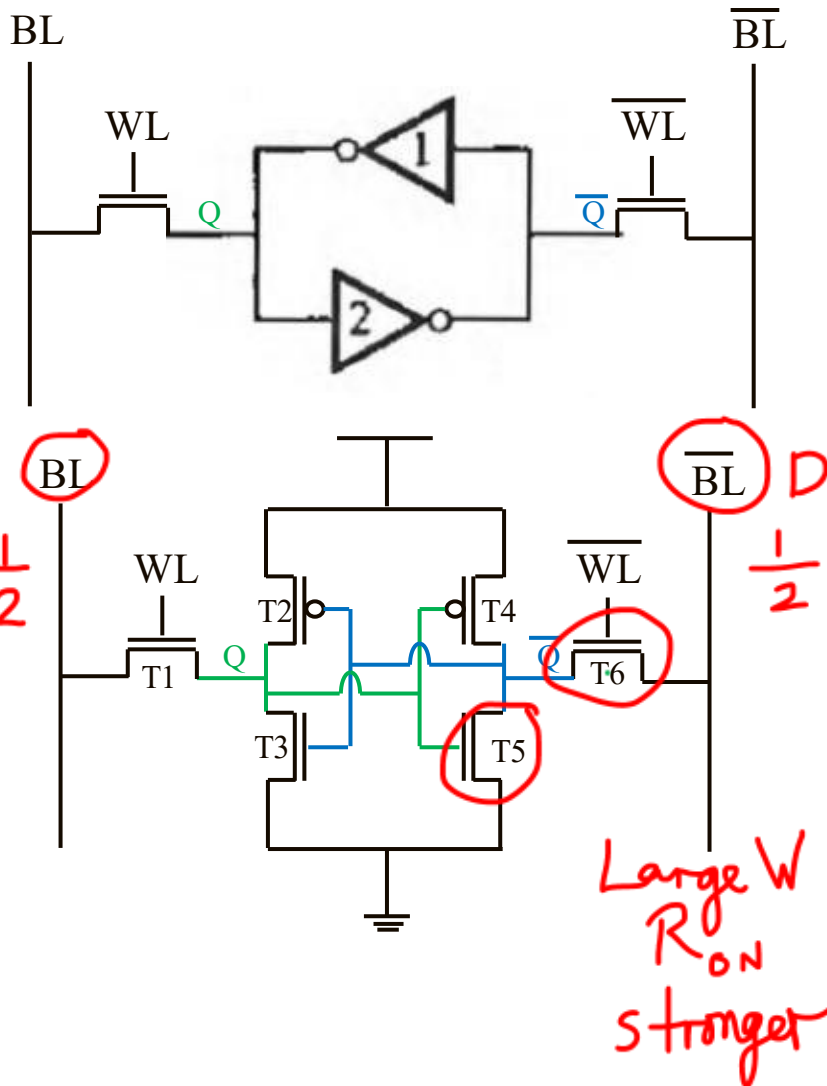


- Ram is organized in 2-D grid
 - Horizontal D is Word (memory address)
 - Vertical D is Bit
 - This is a 4x4 memory
 - 4 words by 4 bits
- Word Line (WL) activates row
 - Connects each memory bit in the word to its respective bit line
 - Read – memory bit sets entire bit line
 - Write – bit line sets the activated memory bits

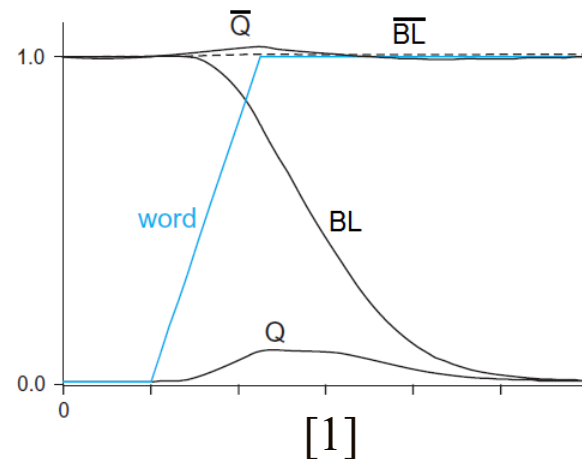
SRAM Cell



SRAM - Read

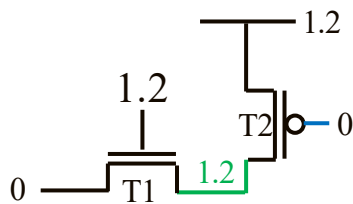


- Bit-line access controlled by NMOS pass transistor
 - Degraded, slow logic 1 pull-up
- One side (BL, WL) is read port
- Other side ($\overline{BL}$, $\overline{WL}$) is write port
- To Read
 - Set WL high
 - Q inverter pulls BL to full logic 0 or 1
 - T2, T3 must overpower T1, or stored data is destroyed
 - $W_{T2,T3} > W_{T1}$

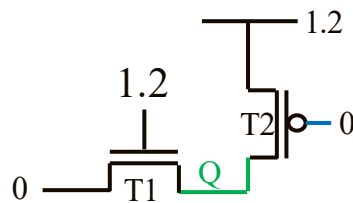


SRAM - Read

- Example – read 1
- Initial condition



- Ensure Q does not drop below V_{IH}



$$W_{T2} = 5W_{T1}$$

$$L = 100\text{nm}$$

Assume T2 linear, T1 saturated

$$\mu_P \frac{W_P}{L} \cancel{C_{ox}} \left[(V_{SG} - |V_{TP}|) (V_{DD} - V_Q) - \frac{(V_{DD} - V_Q)^2}{2} \right] \frac{1}{1 + \frac{V_{DD} - V_Q}{E_c L}}$$

$$= V_{sat} \frac{W_N \cancel{C_{ox}} (V_{GS} - V_{TN})^2}{V_{GS} - V_{TN} + E_c L}$$

$$\frac{70.5 \cdot}{1 + \frac{1.2 - V_Q}{2.4}} \left[(1.2 - 0.4)(1.2 - V_Q) - \frac{(1.2 - V_Q)^2}{2} \right]$$

$$= \frac{8 \cdot 10^6 \cdot 10^{-9} \cdot (1.2 - 0.4)^2}{1.2 - 0.4 + 0.6}$$

$$V_Q = 1.05\text{V}$$

SRAM - Read

$$V_a = 1.05V$$

NMOS check

$$V_{DSN} = 1.05V \quad \text{Sat if } V_{DS} \geq V_{GS} - V_T$$
$$1.05 \geq 1.2 - 0.4$$
$$1.05 \geq 0.8 \quad \checkmark$$

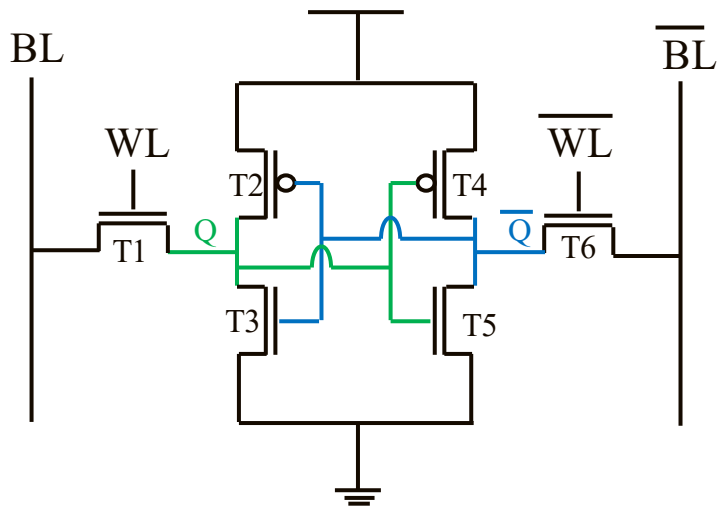
PMOS check

$$V_{SDP} = 1.2 - 1.05$$
$$V_{SDP} = 0.15V$$
$$V_{SD}(\text{sat}) = \frac{(V_{SG} - |V_{TP}|)E_{cPL}}{V_{SG} - |V_{TP}| + E_{cPL}} = \frac{(1.2 - 0.4)2.4}{1.2 - 0.4 + 2.4}$$
$$V_{SD}(\text{sat}) = 0.6V$$

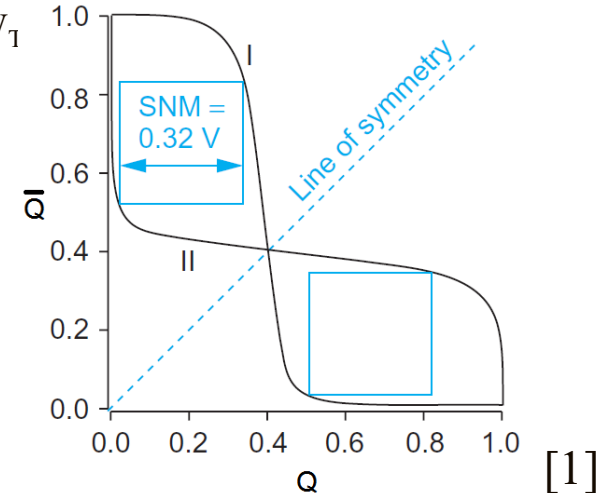
Linear if

$$V_{SDP} \leq V_{SD}(\text{sat})$$
$$0.15V \leq 0.6V \quad \checkmark$$

SRAM - Write

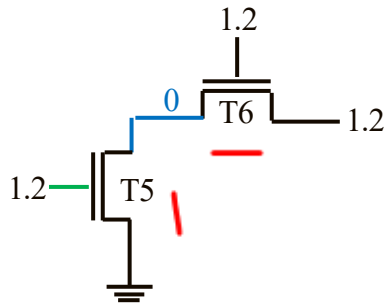


- One side (BL, WL) is read port, other side ($\overline{\text{BL}}$, $\overline{\text{WL}}$) is write port
- To write
 - Set both $\overline{\text{WL}}$ and WL high
 - $\overline{\text{BL}}$ overpowers T4, T5 inverter to pull $\overline{\text{Q}}$ to $\overline{\text{BL}}$ voltage
- Remember $W_{T2,T3} > W_{T1}$ to support read operations on BL, WL side
- For write, T1 must overpower T2, T3, or state cannot change
 - Not possible since port is designed NOT to lose state on read
 - Design T4, T5, T6 for write
 - T6 must overpower T4, T5
 - $W_{T6} > W_{T1}$



SRAM - Write

- Example – write 1
- Initial condition



With W_{T5} only $2 \times W_{T6}$, this will be close.

Estimate: $1.2 \frac{W_6}{W_5 + W_6} = 0.8$, $V_{DS} > 0.4V$, assume both saturated

$$I_{DSS} = I_{DS6}$$

$$\frac{\cancel{V_{sat}} \cancel{W_{T5}} \cancel{C_{ox}} (V_{GSS} - V_{TN})^2}{V_{GSS} - V_{TN} + E_c L} = \frac{\cancel{V_{sat}} \cancel{W_{T6}} \cancel{C_{ox}} (V_{GSS} - V_{TN})^2}{V_{GSS} - V_{TN} + E_c L}$$

$$\frac{(1.2 - 0.4)^2}{1.2 - 0.4 + 0.6} = \frac{2 \cdot (1.2 - V_a - 0.4)^2}{1.2 - V_a - 0.4 + 0.6}$$

$$V_a = 0.3V \text{ fails!}$$

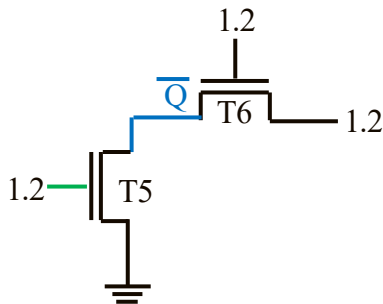
$$V_{IH} \approx 0.7V$$

For success, $W_{T6} \approx 20W_{T5}$

$$W_{T6} = 2W_{T5}$$

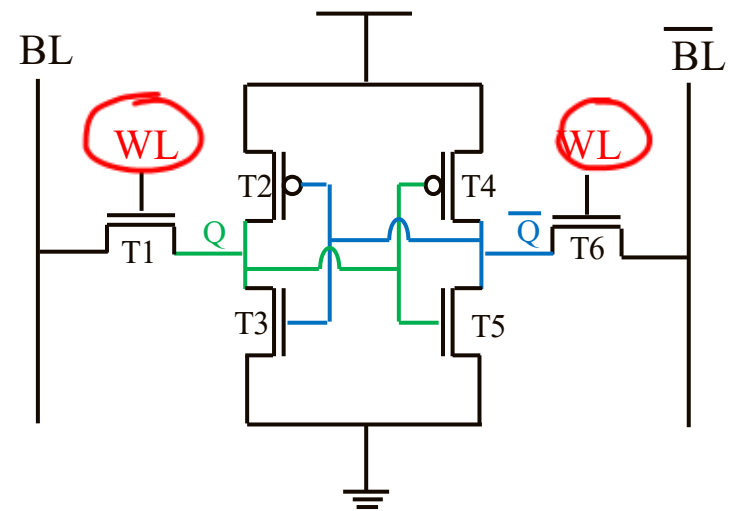
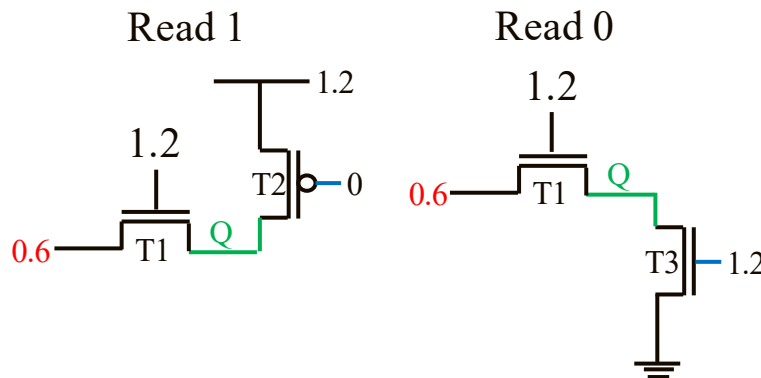
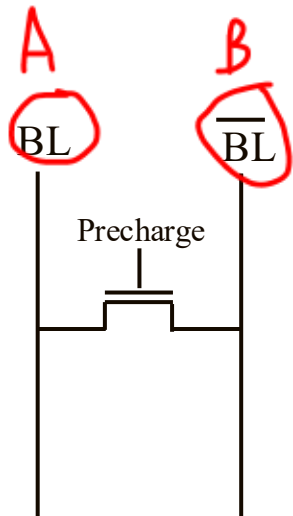
$$L = 100nm$$

- Ensure $\bar{Q}$ does exceed V_{IH}



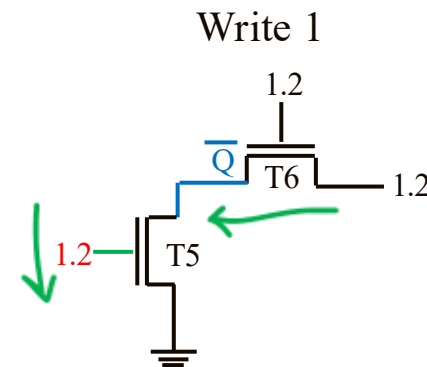
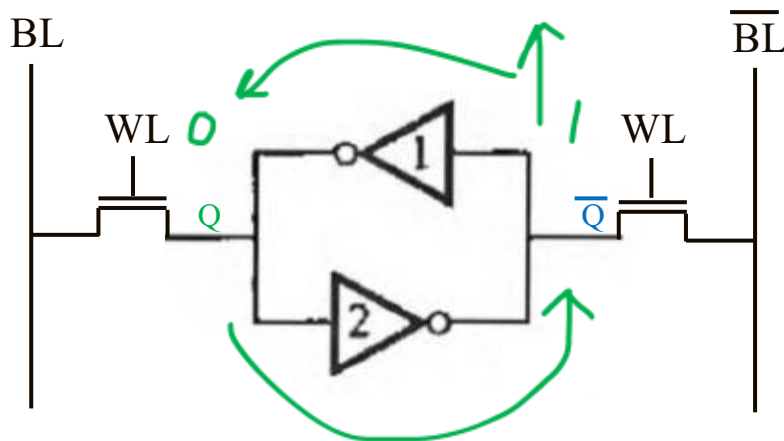
Pre-charging

- As seen previously, for write, excessively large W_{T6} is required
 - Read port encounters same issue for read 0
- Solution – Precharging
- Set both BL and $\overline{BL}$ to $V_{DD}/2$ prior to read
- Can now eliminate $\overline{WL}$, balance transistor sizing



Pre-charging

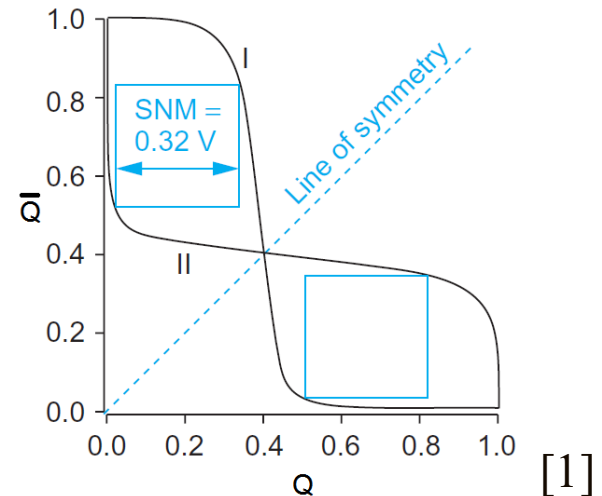
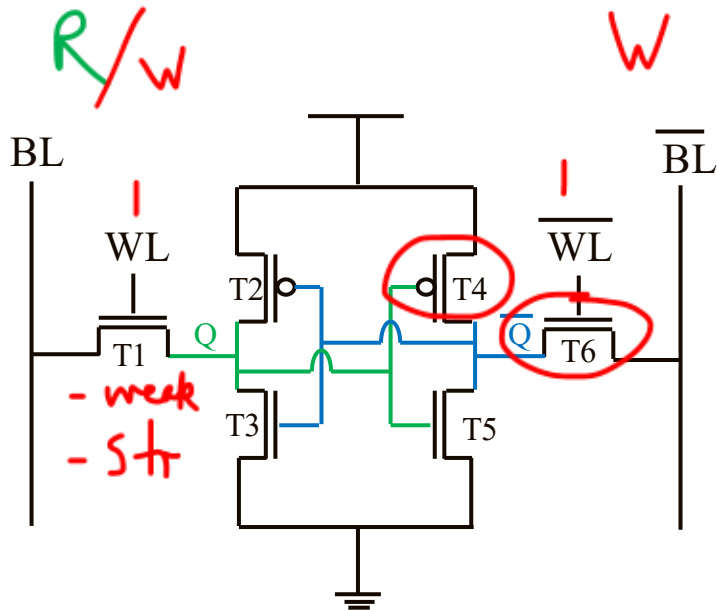
- For write, second inverter now assists substantially
- Opposing transistor will now begin to turn off prior to reaching V_{IH}



V_{GS} now declines when V_{IL} is surpassed on $\overline{Q}$ (SRAM is now symmetric and changes in Q and $\overline{Q}$ mirror each other).

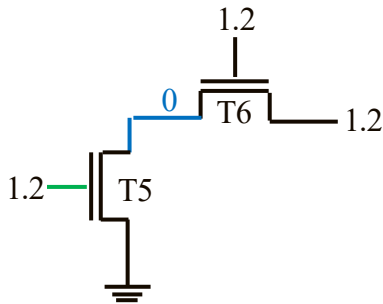
SRAM - Write

- One side (BL, WL) is read port, other side ($\overline{\text{BL}}$, $\overline{\text{WL}}$) is write port
- To write
 - Set both $\overline{\text{WL}}$ and WL high
 - $\overline{\text{BL}}$ overpowers T4, T5 inverter to pull $\overline{\text{Q}}$ to $\overline{\text{BL}}$ voltage
- Remember $W_{T2,T3} > W_{T1}$ to support read operations on BL, WL side
- For write, T1 must overpower T2, T3, or state cannot change
 - Not possible since port is designed NOT to lose state on read
 - Design T4, T5, T6 for write
 - T6 must overpower T4, T5
 - $W_{T6} > W_{T4,T5}$

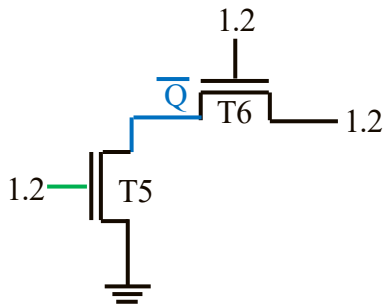


SRAM - Write

- Example – write 0
- Requires setting $\overline{Q}$ to 1
- Initial condition:



- Ensure $\overline{Q}$ does exceed V_{IH}



With W_{T6} only $2 \times W_{T5}$, this will be close.

Estimate: $1.2 \frac{W_6}{W_5 + W_6} = 0.8$, $V_{DS} \geq 0.4V$, assume both saturated

$$I_{DSS} = I_{DS6}$$

$$\frac{V_{sat} \cancel{W_{T5}} \cancel{C_{ox}} (V_{GS5} - V_{TN})^2}{V_{GS5} - V_{TN} + E_c L} = \frac{V_{sat} \cancel{W_{T6}} \cancel{C_{ox}} (V_{GS6} - V_{TN})^2}{V_{GS6} - V_{TN} + E_c L}$$

$$\frac{(1.2 - 0.4)^2}{1.2 - 0.4 + 0.6} = \frac{2 \cdot (1.2 - V_a - 0.4)^2}{1.2 - V_a - 0.4 + 0.6}$$

$$V_a = 0.3V \text{ fails!}$$

$$V_{IH} \approx 0.7V$$

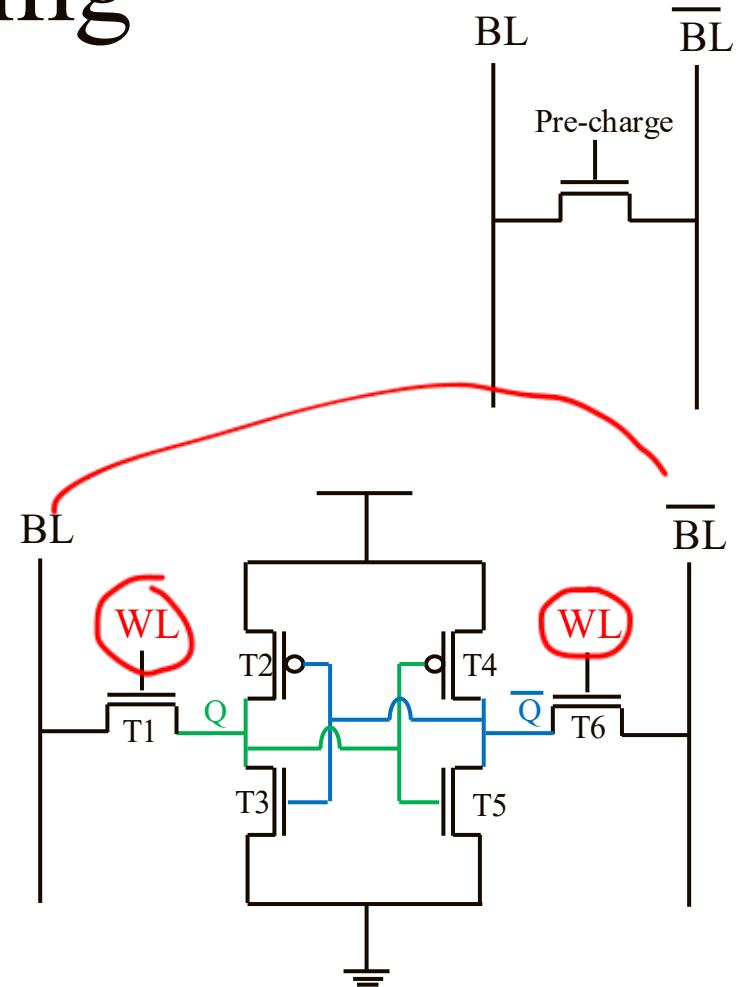
For success, $W_{T6} \approx 20W_{T5}$

$$W_{T6} \neq 2W_{T5}$$

$$L = 100nm$$

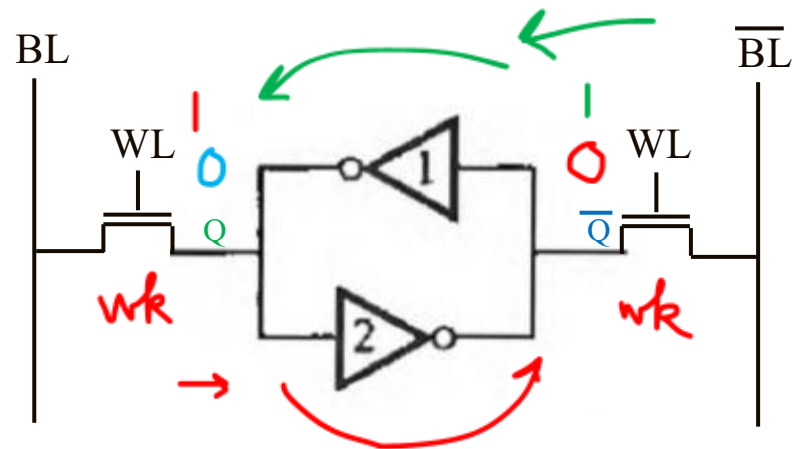
Pre-charging

- As seen previously, for write, excessively large W_{T6} is required
- Solution – Pre-charging
- Two methods:
 - Set both BL and $\overline{BL}$ to $V_{DD}/2$ prior to read
 - Set both BL and $\overline{BL}$ to V_{DD} prior to read
- Both methods eliminate slow logic 1 read issue
 - $V_{DD}/2$ method makes accidental write on read impossible
 - V_{DD} method has accidental write possible, but requires write through degraded V_{GS} pass transistor
- Can now eliminate $\overline{WL}$, balance transistor sizing



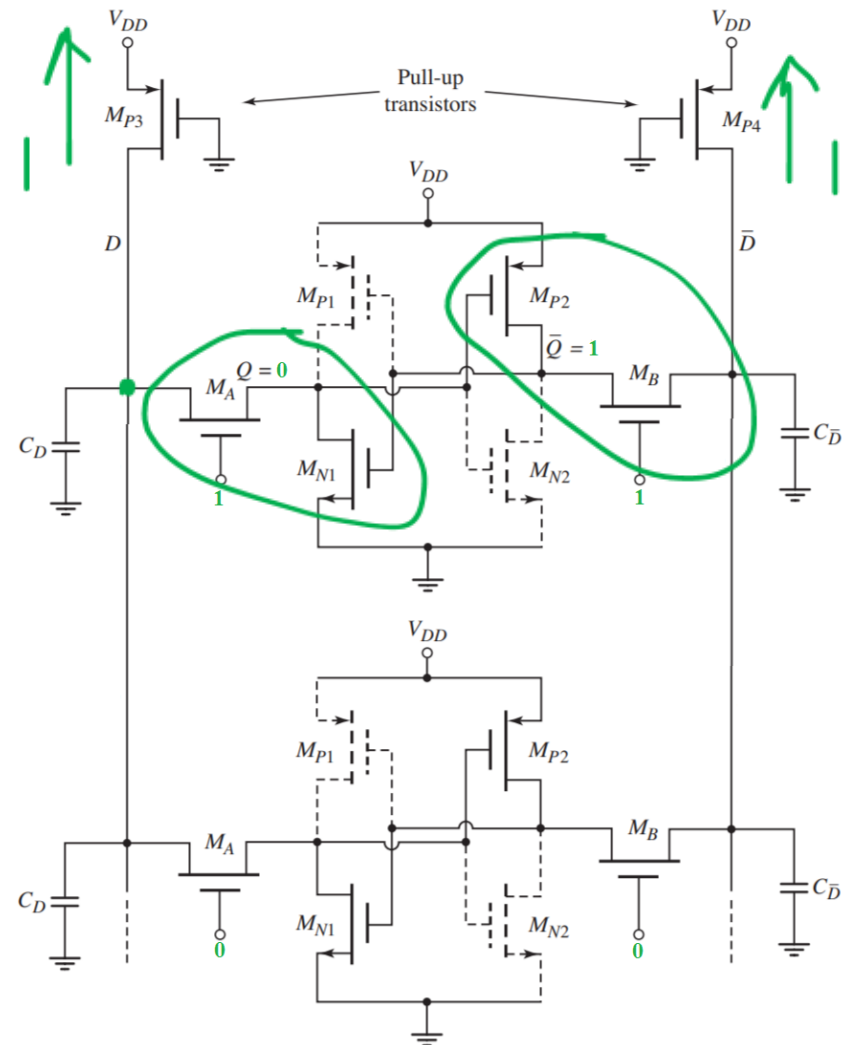
Pre-charging

- For write, both Q and $\bar{Q}$ ports are writable
- This allows us to bypass the weak pull-up capability of the pass transistor
 - For logic 0, write 0 to Q
 - For logic 1, write 0 to $\bar{Q}$ } (green bracket)
- In practice, write to both ports
 - For logic 0, write 0 to Q , 1 to $\bar{Q}$
 - For logic 1, write 0 to $\bar{Q}$, 1 to Q



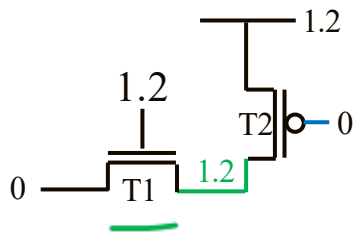
SRAM – Read 1 (V_{DD} Pre-charge)

- Issue – Read of logic 1 is weak
 - Maximum pull-up capability to $V_{DD} - V_T$, for about 0.8V maximum.
- Issue – Read of logic 1 can pull Q low due to strong pass transistor
- One solution – include external pull-up

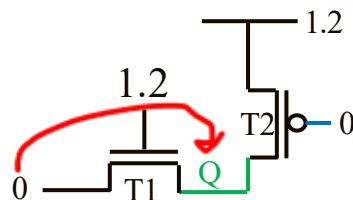


SRAM - Write

- Example – write 0
- Write 0 to Q, 1 to $\bar{Q}$
 - 0 to Q dominates
- Initial condition



- Ensure Q does not drop below V_{IH}



LPF

$$W_{T2} = W_{T1} !$$

$$L = 100\text{nm}$$

$Q = \text{Low expected, } V_{DS2} \text{ is high, } T2 \text{ saturated}$
 $V_{DS1} \text{ is low, } T1 \text{ is linear}$

$$\frac{\mu_n \cancel{C_{ox}} \frac{W}{L}}{1 + \frac{V_{DS1}}{E_{eff}}} \left[(V_{GS1} - V_{T1}) V_{DS1} - \frac{V_{DS1}^2}{2} \right] = \frac{W_p \cancel{C_{ox}} (V_{SG2} - |V_{T2}|)^2 \cdot V_{SD2}}{V_{SG2} - |V_{T2}| + E_{eff}}$$

$$\frac{270}{1 + \frac{V_Q}{0.6}} \cdot \left[(1.2 - 0.4) V_Q - \frac{V_Q^2}{2} \right] = \frac{10^{-9} (1.2 - 0.4)^2 \cdot 8 \cdot 10^6}{1.2 - 0.4 + 2.4} \quad \checkmark$$

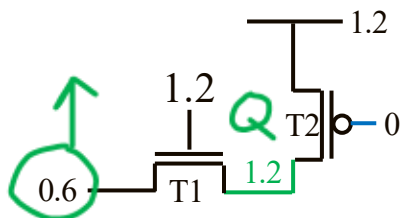
$$V_Q = 0.09\text{V}$$

0.1V

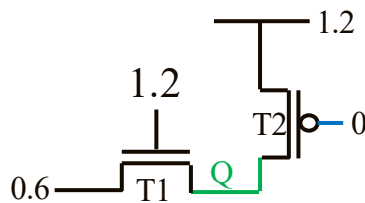
$V_Q < V_{IL}$, write succeeds

SRAM – Read ($V_{DD}/2$ Pre-charge)

- Example – read 1
- Initial condition



- Ensure Q does not drop below V_{IH}



~~$W_{T2} = W_{T1}$~~
 $L = 100\text{nm}$

$V_{SD}(\text{sat})$ of T2 is 0.6V. This cannot be exceeded, T2 must be linear

$$V_{DS}(\text{sat}) \text{ of } T1 = \frac{(V_{GS1} - V_{T1}) E_c L}{V_{GS1} - V_{T1} + E_c L} = \frac{(0.6 - 0.4)(0.6)}{0.6 - 0.4 + 0.6} = 0.15\text{V}$$

To check if V_Q drops below V_{IH} of 0.7V, T1 must be linear.

$$I_{DS1} = I_{SD2}$$

$$\frac{\mu_n C_{ox} \frac{W_1}{L}}{1 + \frac{V_{DS1}}{E_c L}} \left[(V_{GS1} - V_{T1}) V_{DS1} - \frac{V_{DS1}^2}{2} \right] = \frac{\mu_p C_{ox} \frac{W_2}{L}}{1 + \frac{V_{SD2}}{E_c L}} \left[\dots \right]$$

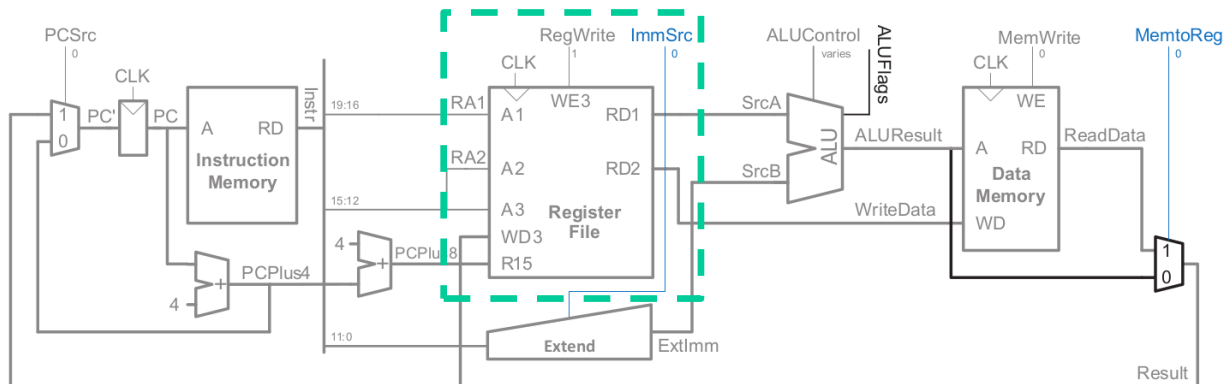
$$\frac{270}{1 + \frac{V_Q - 0.6}{0.6}} \cdot \left[(1.2 - 0.6 - 0.4)(V_Q - 0.6) - \frac{(V_Q - 0.6)^2}{2} \right] = \frac{70 \cdot 1}{1 + \frac{1.2 - V_Q}{2.4}} \left[(1.2 - 0.4)(1.2 - V_Q) - \frac{(1.2 - V_Q)^2}{2} \right]$$

$V_Q = \text{imaginary number!}$

$7W_{T2} = W_{T1}$ yields $V_Q \approx 0.7\text{V}$

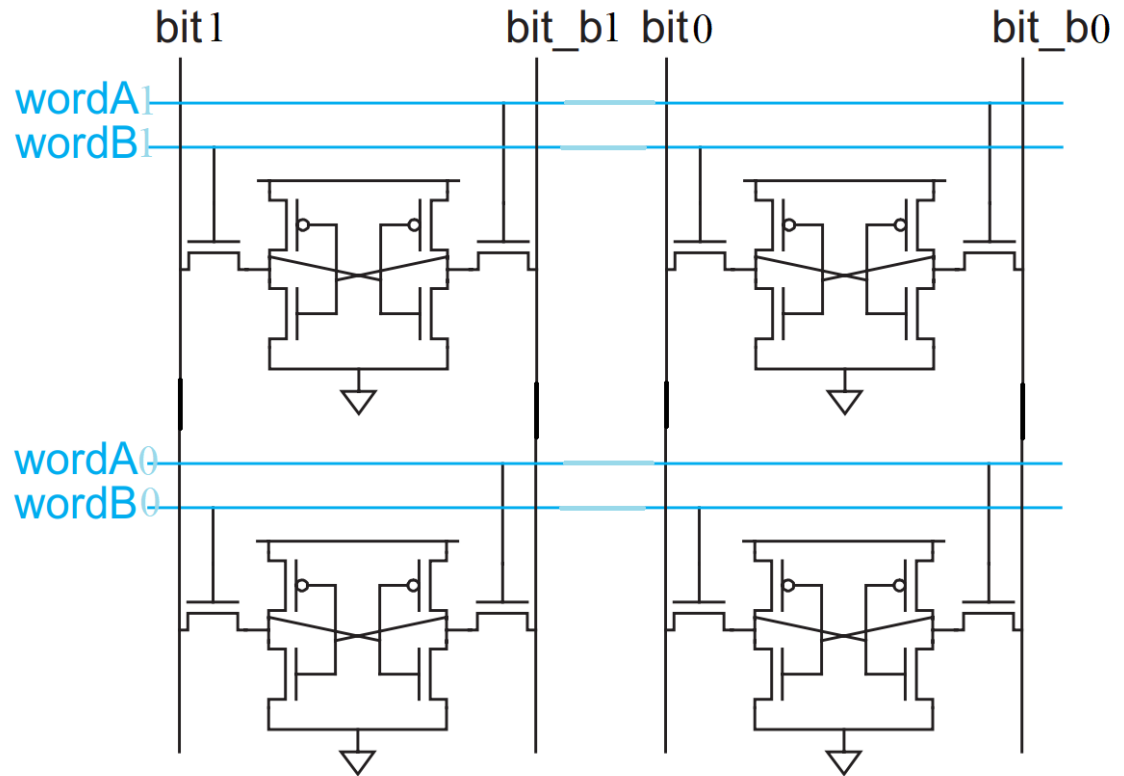
Multi-Port RAM

- SRAM typically used in cache memory, register file.
 - Cache memory typically only 1 access needed per cycle.
 - Register file requires many accesses per cycle.
- Activating a word-line for read blocks out all other reads from RAM array
 - Inadequate for register file.
 - Multi-port memory required.
- Below – ARMv4 2-port register file from CHD lab 5.



Simple Multi-Port RAM

- For read, activate wordA or wordB
 - Results in both bit-lines being read independently
- For write, must activate both wordA and wordB



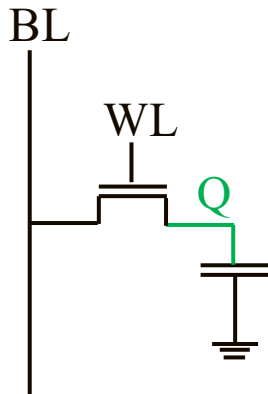
SRAM

- Static
 - Memory data is stored indefinitely as long as power is maintained
 - Non-destructive read-out
 - Do not need to re-write data after read
- Lower logic density
 - 6 transistors
 - 2 bit-lines per cell
- Faster
 - Inverter provides continual current drive on output bit-lines

DRAM

- Dynamic
 - memory data is lost over time even if power is maintained
 - Capacitors slowly drain
 - Must be refreshed every $\sim 10\mu\text{s}$
 - Destructive read-out
 - Must re-write data back
- Higher logic density
 - Just 1 transistor
 - Capacitor is stacked below transistor
 - 1 bit line, word line per cell
- Slower
 - Capacitor only provides a small burst of charge to/from the bit-line

DRAM



Read

- DRAM capacitor charge can only slightly shift bit line voltage
- BL is precharged to $V_{DD}/2$
- WL is activated
- Logic 0
 - Q is at GND voltage
 - BL shifts below $V_{DD}/2$
- Logic 1
 - Q is at $V_{DD} - V_t$ voltage
 - BL shifts above $V_{DD}/2$
- Sense amplifier
 - Logic 1 of V_{DD} , logic 0 of GND is required
 - Sense amplifier compares BL to $V_{DD}/2$, and pulls voltage to V_{DD} or GND depending on BL offset voltage from bit read
- Must write data back after read

Write

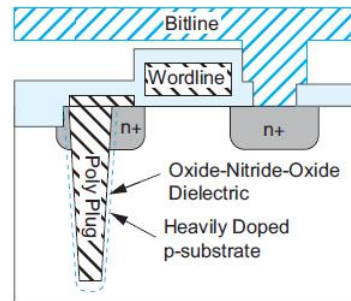
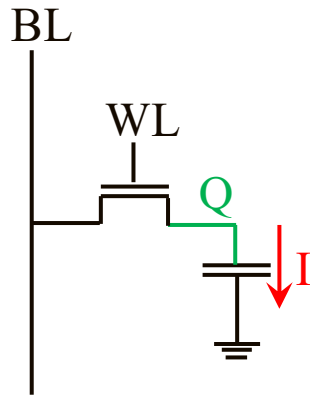
Logic 1

Set BL to logic 1
 Activate WL
 Q is set to $V_{DD} - V_t$

Logic 0

Set BL to logic 0
 Activate WL
 Q is set to GND

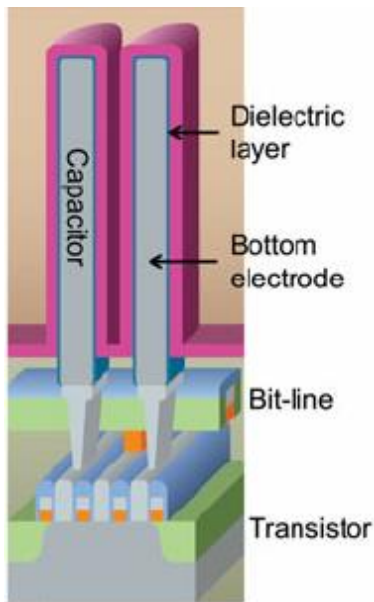
DRAM Capacitor



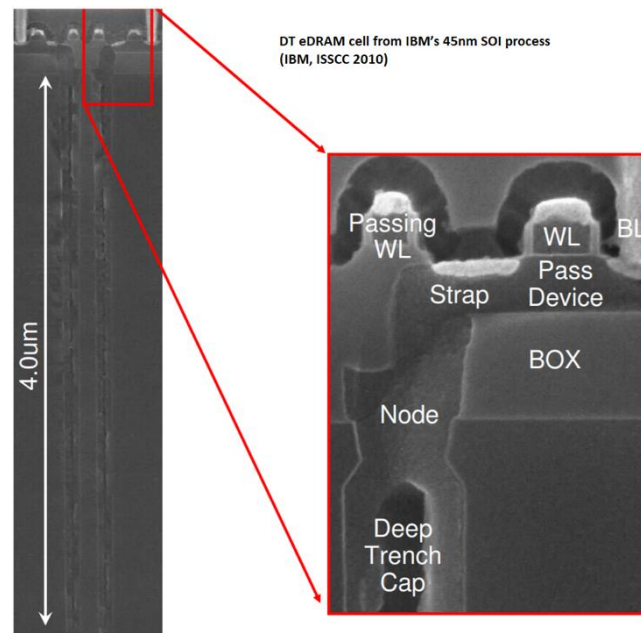
[1]

- $C \sim \epsilon_{ox} \text{Area}/d$
- Capacitor is similar to a MOSFET gate capacitor
 - Thin gate oxide (2nm to 100nm) results in high capacitance
- DRAM capacitor leaks over time
- Even with no activity, must be refreshed every 10-100 μs
- On modern DRAM capacitor is stacked over top of transistor
 - In CMOS logic, this is where we want interconnect wires
 - Integration of DRAM on CMOS processor die very difficult
 - Typically when fast DRAM is desired, a separate die on the same package is used
 - Only IBM (invented DRAM) sold processors with on-die DRAM

DRAM – Stacked & Trench Storage Capacitors



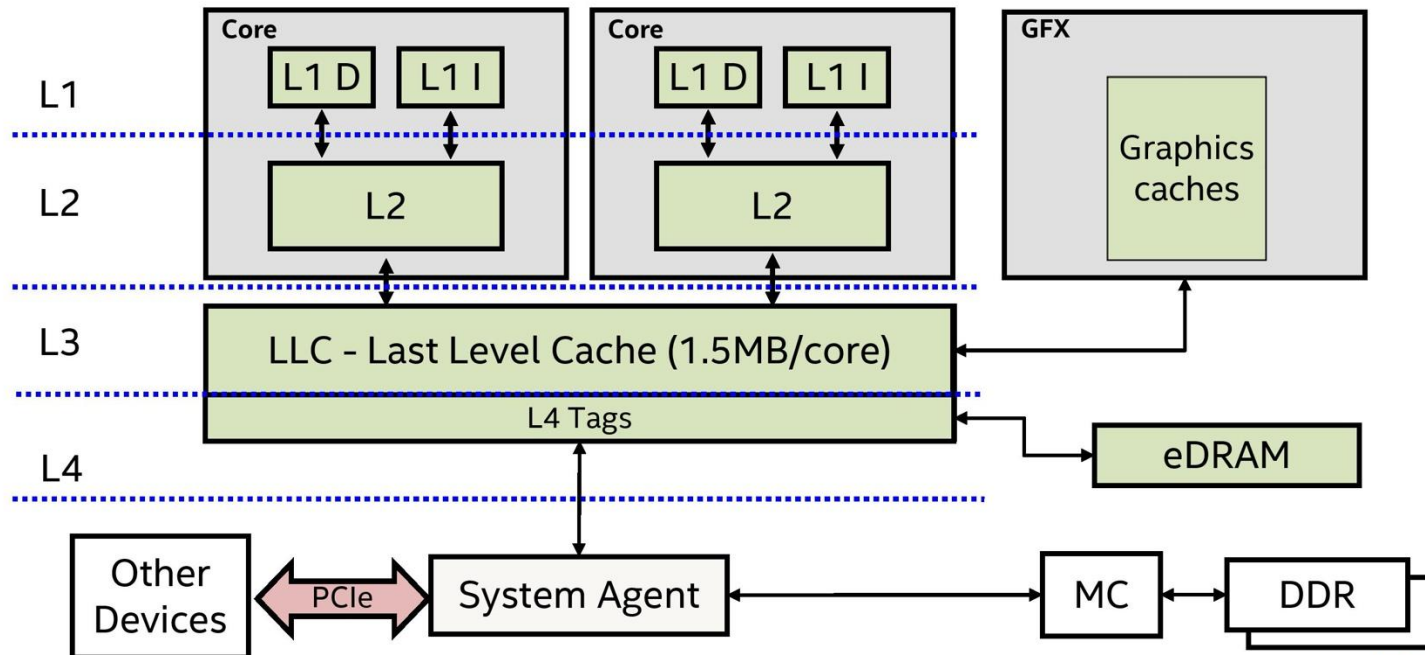
Stacked – Cap over transistor [2]



Trench – Cap under transistor

eDRAM

eDRAM Based Cache



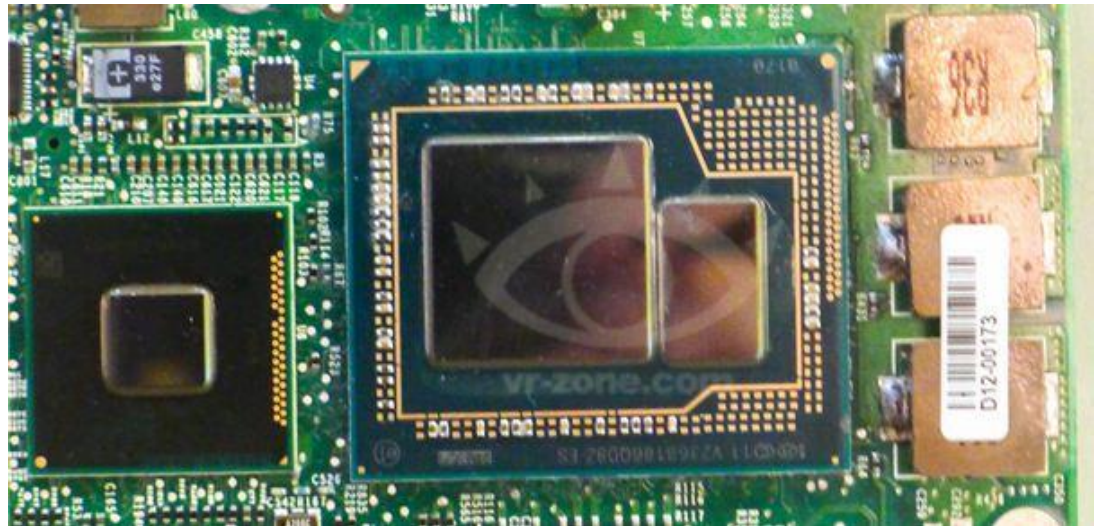
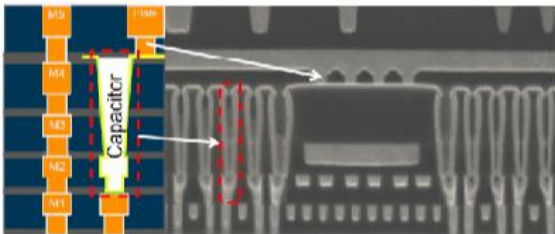
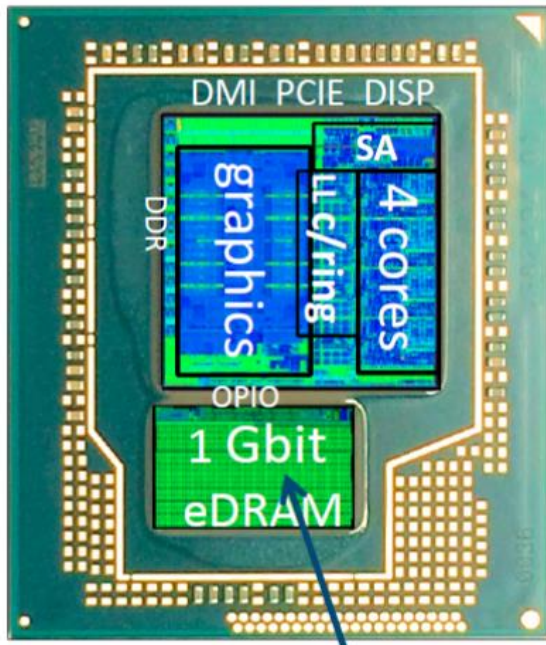
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Intel Next Generation Microarchitecture Code Name Skylake

IDF15
INTEL DEVELOPER FORUM

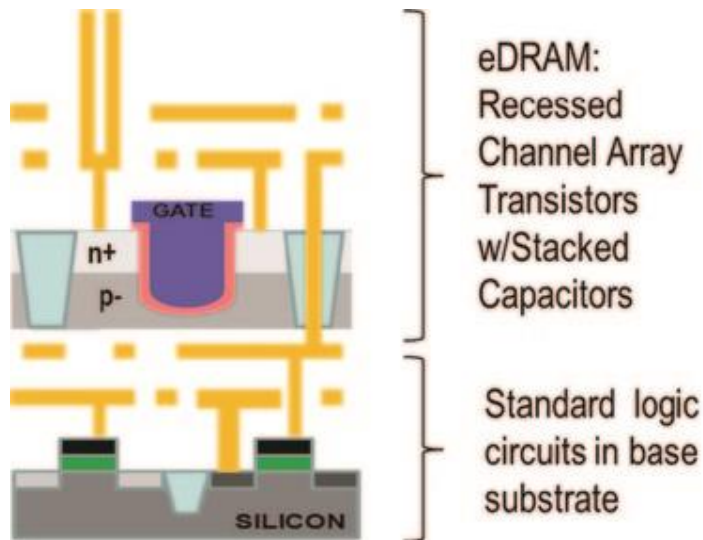
eDRAM

Intel Haswell GT3e



3D Embedded DRAM

Cell Structure

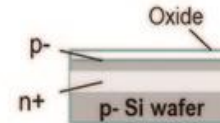


Fabrication Process

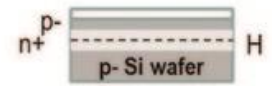
(a) Build transistors and interconnects for bottom wafer of 3D stack



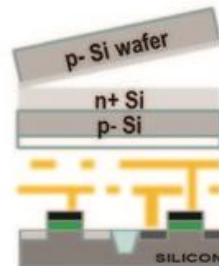
(b) Using new wafer, construct dopant regions and activate at -1000°C



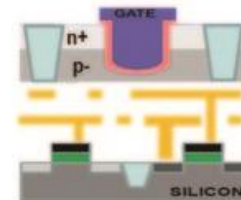
(c) Implant hydrogen for Ion-Cut
↓ ↓ ↓ ↓ ↓



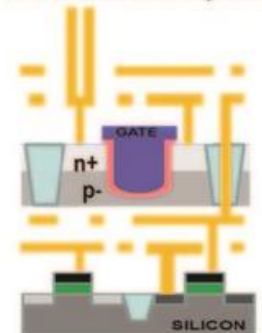
(d) Flip, bond, cleave, polish



(e) Form RCJLTs using low temperature steps such as etch and deposition



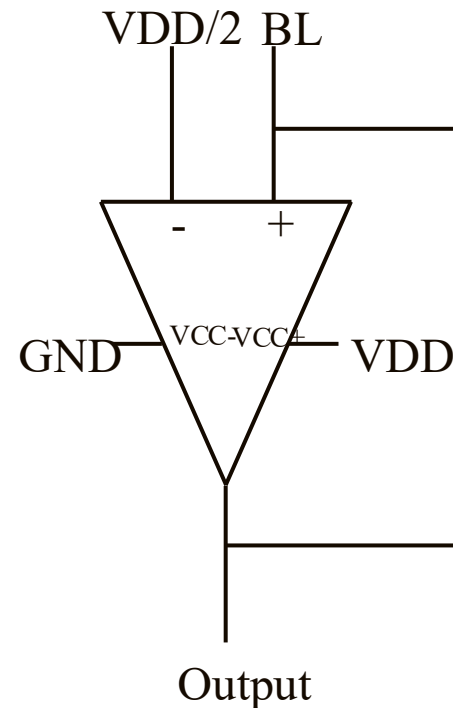
(f) Complete wires & through-vias. Add stacked capacitor.



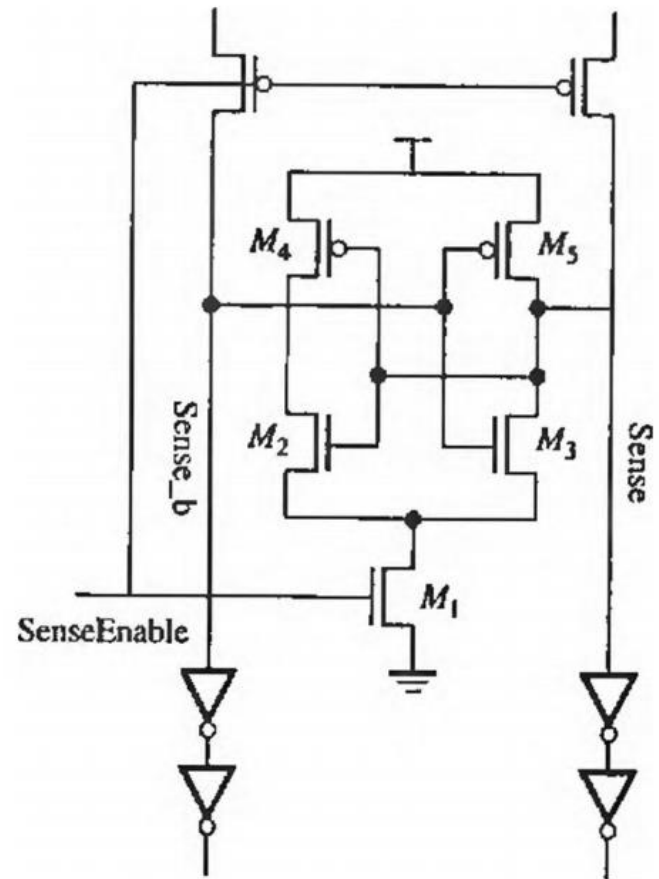
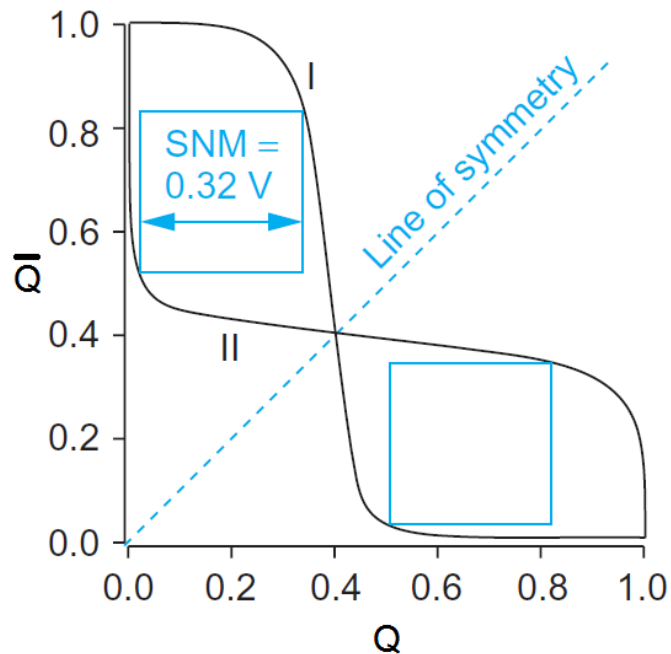
<http://www.monolithic3d.com/3d-embedded-ram.html>

Sense Amplifier

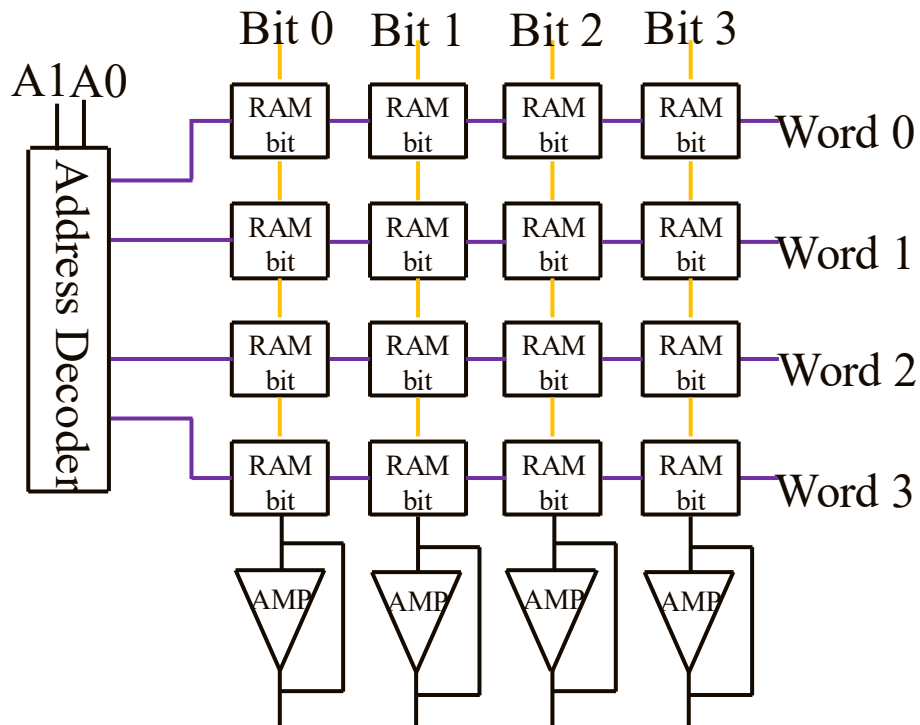
- C_{DRAM} is tiny compared to Bit-Line capacitance
 - Only results in small shift of bit-line voltage
- With DRAM, pulling BL voltage to VDD or GND is required on data read
- This is achieved with a sense amplifier
 - General name of amplifier is a difference amplifier
 - Functions like a comparator
 - Takes the difference between two input voltages, and amplifies the difference on its outputs.
 - In this case, output loops back to the input
 - Positive feedback
- While not required, SRAM also uses sense amplifier for more speed
- Typically 4 transistors
 - Low gain, but must be small



Sense amplifier

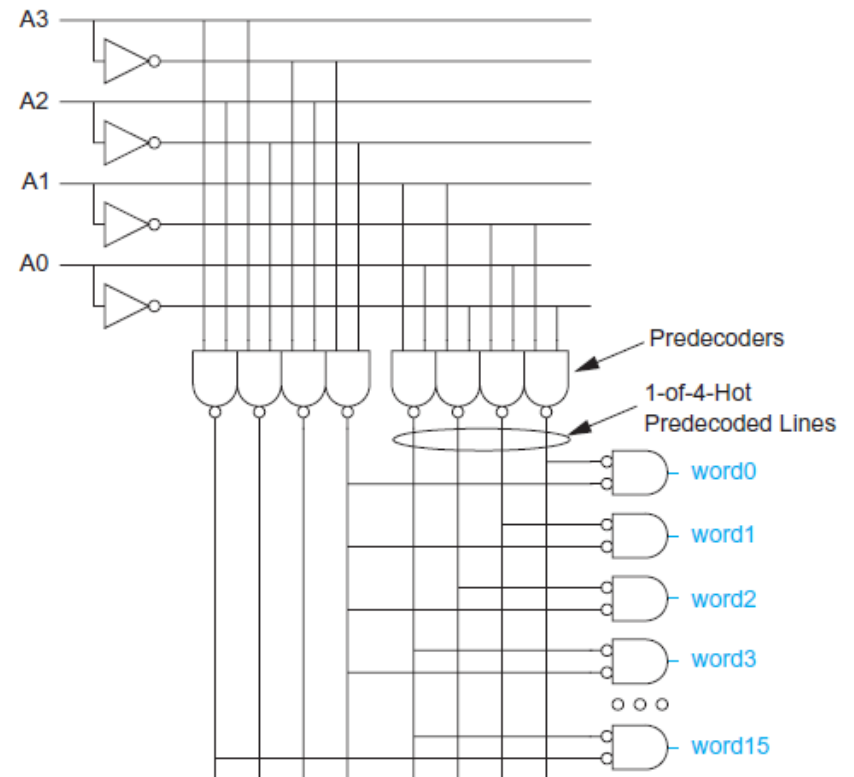
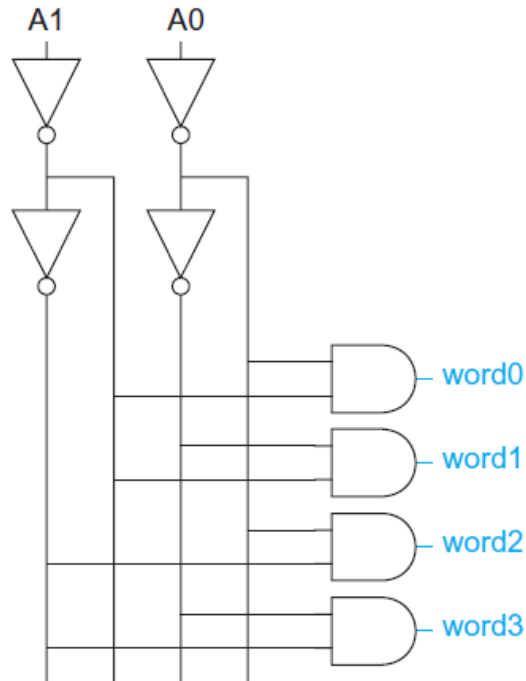


RAM Arrays - Extended



- Address Decoder
 - This 4x4 memory has 2 address bits to access 4 words of memory

Address Decoder



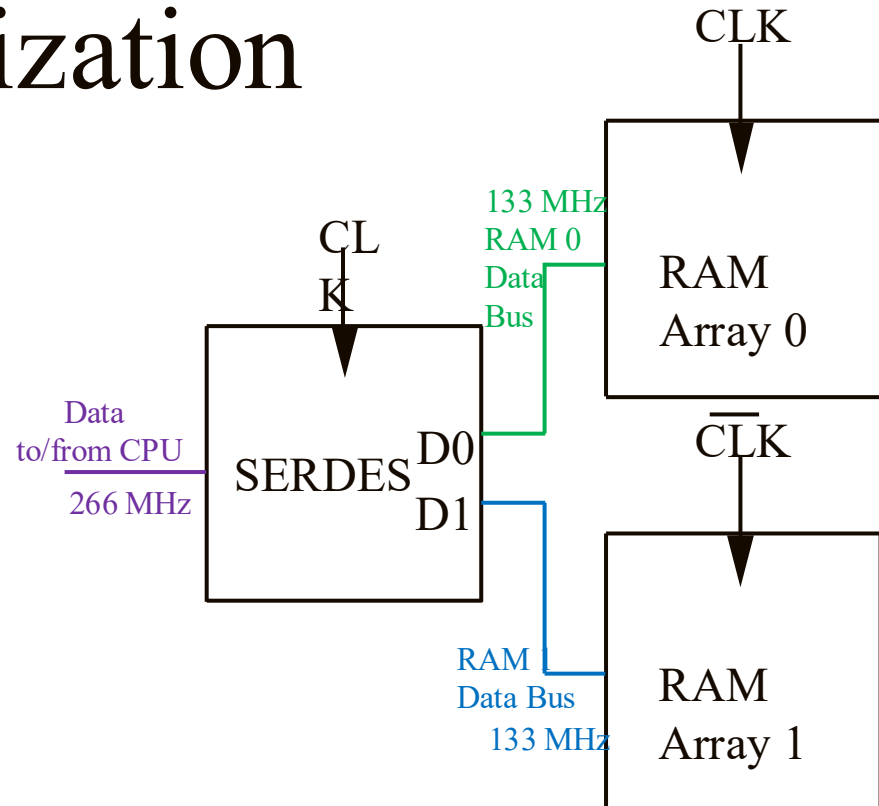
[1]

High Frequency DRAM

- Over past several decades, DRAM access latency has remained unchanged at roughly 100 ns.
- Reason – Reducing size of memory array does not decrease latency
- Despite this, memory frequency continues to increase over time
 - 90's SDR 133 MHz
 - Early 2000's DDR 400 MHz
 - Mid 2000's DDR2 1066 MHz
 - Early 2010's DDR3 2133 MHz
 - Today DDR4 4000 MHz

DRAM Serialization-Deserialization

- DRAM data rate is increased by serializing parallel, interleaved data read/writes to a common data stream.
- Requires SERDES
 - SERIALIZER for read
 - DESERIALIZER for write
 - CLK select Array 0
 - CLK select Array 1
 - Bidirectional, like mux/demux combo
- DDR
 - One array read/write on positive CLK
 - Other array read/write on negative CLK
 - 2 bits returned on every CLK
 - Hence Double Data Rate name
 - Newer DDR standards multiply the CLK to interleave data from/to 4 or 8 separate memory arrays

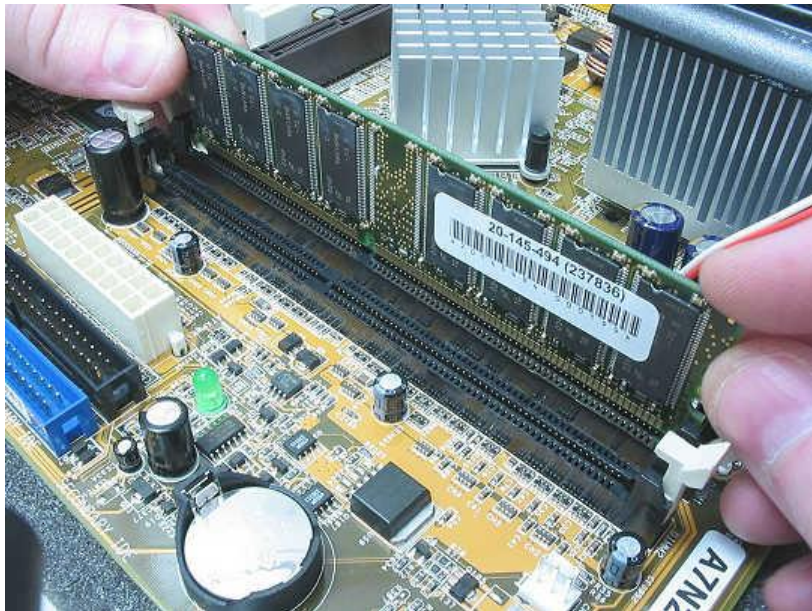


DRAM – Form Factors

DRAM

External from processor

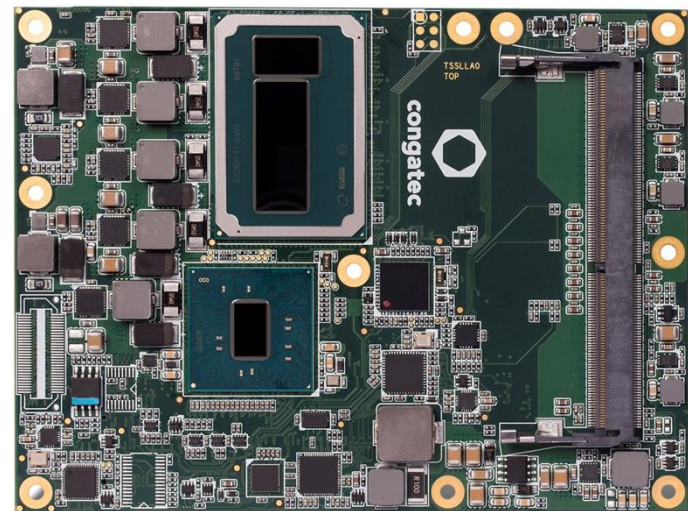
Typically in dedicated memory slots



[4]

Why do processors have the raw silicon exposed while RAM does not?

- eDRAM
 - Embedded on Processor package
 - Located top-center
 - 128 MB eDRAM (22nm process)
 - 4 core Intel Broadwell processor with large integrated graphics (14nm process)



[3]

Heat!



[5]

References

- [1] Weste and Harris CMOS VLSI Design
- [2] https://www.researchgate.net/figure/Schematic-diagrams-of-a-DRAM-cells-which-consist-of-a-cell-transistor-and-capacitor_fig1_258797946
- [3] https://images.anandtech.com/doci/10281/Congatec_CS170.jpg
- [4] <http://www.mechbgon.com/build/memory1.jpg>
- [5] <https://aroundhomediy.com/article-images/build-amd-ryzen-3700x-computer/build-amd-ryzen-3700x-computer-07.jpg>